



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Mid Exam, Fall 2025

CSE 4611/CSI 411: Compiler Design/Compiler

Total Marks: 30 Duration: 1 hour 30 Minutes

Answer all the questions. Figures in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

- 1 a) Show the corresponding inputs and outputs from **Lexical analyzer** to **Code Optimizer** phase of compiler based on the following expression where **a** and **b** are **float** types and other are **int** types.[need to show intermediate steps] 6

$$x = a^{2*(b+c)}/b$$

- b) Two Computer Science undergrad students were discussing about code efficiency. They were comparing their code with each other. Even though both of them solved the problem in differently but had different number of lines the compiler generated same number of line as machine code. Which component of a compiler caused this? Show you reasoning 2

- 2 Draw Syntax Tree for the following expression. 3

$$2 * ((a^3)^{2^{(b/c)}})/2/6 - (b/p^3) * a * 2 * (6 - 2 * c)$$

- 3 Prove that the following expression “((7/y)((x)/y))” can be represent by the grammar given below. 4

$$P \rightarrow (PS) \mid (SP) \mid /RP \mid Q \mid \epsilon$$

$$Q \rightarrow SP \mid R \mid x \mid y \mid \epsilon$$

$$R \rightarrow *RQ \mid ^RQ \mid Q \mid \epsilon$$

$$S \rightarrow (P) \mid Q \mid 3 \mid 7 \mid \epsilon$$

- 4 Write down the optimized three address representation of the following expression. 4

$$Z = (a+(c*d/4)^{4/b})*2/(e^{2^{2-x/a}}/(c*d/4)^a)$$

5. Convert the following infix expression to a prefix expression. 4

$$(a + (b \wedge c + d / (2 * a)^6) / 2 + 3) + (2^e \% 2) / 2^3 / a^a a^a$$

6. Convert the following infix expression to a postfix expression. 4

$$2 \wedge 6 + 5 * (8 / 4 \wedge 2 / 4 * a) \wedge 7 * 6 \wedge (a + 6 * 2) \wedge 6 - 7 * 4$$

7. Evaluate the following expression. [show calculations] 3

$$+ - - / \wedge + 2 \ 3 \ 4 + 4 \ 4 * 3 \ 2 * / 4 \ 2 \ 8 / * 8 + 2 \ 2 \ 6$$